Introduction

Purpose of this research is about how the students' grades are effected by the kind of food, eating habit and other personal lifestyle. And vice versa, how food effects the students in general.

The dataset is obtained from Mercyhurst University interview with 126 university students and from their responds.

Data itself is raw and uncleaned and I believe this is a good oppotunity to learn about NLP. Data size also fitted within the suggested option. So, the objective of this project is to clean the raw data.

Relevance

For this research project dataset, I obtained it from Kaggle . There are other related dataset regarding with this topic, however, they are the studies about primary, secondary schooler or high school students and not about university students. Since I wanted to do something releated with university students, those datasets are eliminated. There are other datasets about general population with food consumption but the data size is huge. Regarding with food waste dataset, the data itself is too simple with no challenges. So the chosen dataset is the best fit for the project.

Project Background

- Since I am also a university student, I am eager to know how the food consuming have impact on our performance on the school work.
- Dataset provider does the interview out of interest but with that said the data itself is raw and uncleaned which means answers are not being discovered yet. There are some inspiration question stated on the website but they are not answered and also no update is made by the time this project is submitted. I would also be using some of the inspirated question from the dataset provider to find out the interseting pattern and relativeness.
- In the dataset, there are irrelevant questions regarding with the project such as in the heading 'meals_dinner_friend', "what would you serve to friend for dinner?" and etc are eliminated. Since it does not provide any relevance with the objective of this project as in the person here is not the consumer itself but the person he/she treat to.
- Initially, I tried to roughly divide into 3 categories,

1. Demographic
2. Cooking Habits by parents
3. Eating habits after getting into college
4. Other habits affecting the GPA

And some of the questions came from the dataset provider itself since the questions are not being answered or discovered yet.

In [1]:
```
import pandas as pd
df = pd.read_csv('food_coded.csv')
df.head(5)
```

Out[1]:

	GPA	Gender	breakfast	calories_chicken	calories_day	calories_scone	coffee	comfort_food	comfort_food_reasons	comfort_food_reasons_coded	...	soup	sports	thai_food	tortilla_calories	turkey_calories	type_sports	veggies_day	vitamins	waffle_calories	weigl
0	2.4	2	1	430	NaN	315.0	1	none	we dont have comfort	9.0	...	1.0	1.0	1	1165.0	345	car racing	5	1	1315	18
1	3.654	1	1	610	3.0	420.0	2	chocolate, chips, ice cream	Stress, bored, anger	1.0	...	1.0	1.0	2	725.0	690	Basketball	4	2	900	15
2	3.3	1	1	720	4.0	420.0	2	frozen yogurt, pizza, fast food	stress, sadness	1.0	...	1.0	2.0	5	1165.0	500	none	5	1	900	I'm n answerin thi
3	3.2	1	1	430	3.0	420.0	2	Pizza, Mac and cheese, ice cream	Boredom	2.0	...	1.0	2.0	5	725.0	690	NaN	3	1	1315	Not sur 24
4	3.5	1	1	720	2.0	420.0	2	Ice cream, chocolate, chips	Stress, boredom, cravings	1.0	...	1.0	1.0	4	940.0	500	Softball	4	2	760	19

5 rows × 61 columns

Originally, there are a total of 126 rows (students) and 61 columns (informations, 1 column for numbering)

After observing the top 10 rows of the file, the data is pretty messy.

- Most of them are open ended questions which is a bunch of strings with commars.
- NaN values.
- Some data are useful and interesting but they are in unrelated format/datatype
- Some columns are not necessay for this project, for example, in **breakfast** column, showing 2 pictures of 2 different kind of coffee but the answers in either number 1 or 2.

In [3]:
```
df.head()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 125 entries, 0 to 124
Data columns (total 61 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   GPA                           123 non-null    object 
 1   Gender                        125 non-null    int64  
 2   breakfast                     125 non-null    int64  
 3   calories_chicken              125 non-null    int64  
 4   calories_day                  106 non-null    float64
 5   calories_scone                124 non-null    float64
 6   coffee                        125 non-null    int64  
 7   comfort_food                  124 non-null    object 
 8   comfort_food_reasons          123 non-null    object 
 9   comfort_food_reasons_coded    106 non-null    float64
 10  cook                          122 non-null    float64
 11  comfort_food_reasons_coded.1  125 non-null    int64  
 12  cuisine                       108 non-null    float64
 13  diet_current                  124 non-null    object 
 14  diet_current_coded            125 non-null    int64  
 15  drink                         123 non-null    float64
 16  eating_changes                122 non-null    object 
 17  eating_changes_coded          125 non-null    int64  
 18  eating_changes_coded1         125 non-null    int64  
 19  eating_out                    125 non-null    int64  
 20  employment                    116 non-null    float64
 21  ethnic_food                   125 non-null    int64  
 22  exercise                      112 non-null    float64
 23  father_education              124 non-null    float64
 24  father_profession             122 non-null    object 
 25  fav_cuisine                   123 non-null    object 
 26  fav_cuisine_coded             125 non-null    int64  
 27  fav_food                      123 non-null    float64
 28  food_childhood                124 non-null    object 
 29  fries                         125 non-null    int64  
 30  fruit_day                     125 non-null    int64  
 31  grade_level                   125 non-null    int64  
 32  greek_food                    125 non-null    int64  
 33  healthy_feeling               125 non-null    int64  
 34  healthy_meal                  124 non-null    object 
 35  ideal_diet                    124 non-null    object 
 36  ideal_diet_coded              125 non-null    int64  
 37  income                        124 non-null    float64
 38  indian_food                   125 non-null    int64  
 39  italian_food                  125 non-null    int64  
 40  life_rewarding                124 non-null    float64
 41  marital_status                124 non-null    float64
 42  meals_dinner_friend           122 non-null    object 
 43  mother_education              122 non-null    float64
 44  mother_profession             123 non-null    object 
 45  nutritional_check             125 non-null    int64  
 46  on_off_campus                 124 non-null    float64
 47  parents_cook                  125 non-null    int64  
 48  pay_meal_out                  125 non-null    int64  
 49  persian_food                  124 non-null    float64
 50  self_perception_weight        124 non-null    float64
 51  soup                          124 non-null    float64
 52  sports                        123 non-null    float64
 53  thai_food                     125 non-null    int64  
 54  tortilla_calories             124 non-null    float64
 55  turkey_calories               125 non-null    int64  
 56  type_sports                   99 non-null     object 
 57  veggies_day                   125 non-null    int64  
 58  vitamins                      125 non-null    int64  
 59  waffle_calories               125 non-null    int64  
 60  weight                        123 non-null    object 
dtypes: float64(20), int64(27), object(14)
memory usage: 59.7+ KB
```

Checking the datatype of each column to make sure the datatype in order to do data cleaning in the following.

In [4]:
```
df = df.drop_duplicates()
df
```

Out[4]:

	GPA	Gender	breakfast	calories_chicken	calories_day	calories_scone	coffee	comfort_food	comfort_food_reasons	comfort_food_reasons_coded	...	soup	sports	thai_food	tortilla_calories	turkey_calories	type_sports	veggies_day	vitamins	waffle_calories	we
0	2.4	2	1	430	NaN	315.0	1	none	we dont have comfort	9.0	...	1.0	1.0	1	1165.0	345	car racing	5	1	1315	
1	3.654	1	1	610	3.0	420.0	2	chocolate, chips, ice cream	Stress, bored, anger	1.0	...	1.0	1.0	2	725.0	690	Basketball	4	2	900	
2	3.3	1	1	720	4.0	420.0	2	frozen yogurt, pizza, fast food	stress, sadness	1.0	...	1.0	2.0	5	1165.0	500	none	5	1	900	I'm answ
3	3.2	1	1	430	3.0	420.0	2	Pizza, Mac and cheese, ice cream	Boredom	2.0	...	1.0	2.0	5	725.0	690	NaN	3	1	1315	Not s
4	3.5	1	1	720	2.0	420.0	2	Ice cream, chocolate, chips	Stress, boredom, cravings	1.0	...	1.0	1.0	4	940.0	500	Softball	4	2	760	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
120	3.5	1	1	610	4.0	420.0	2	wine. mac and cheese, pizza, ice cream	boredom and sadness	NaN	...	1.0	1.0	5	940.0	500	Softball	5	1	1315	
121	3	1	1	265	2.0	315.0	2	Pizza / Wings / Cheesecake	Loneliness / Homesick / Sadness	NaN	...	1.0	NaN	4	940.0	500	basketball	5	2	1315	
122	3.882	1	1	720	NaN	420.0	1	rice, potato, seaweed soup	sadness	NaN	...	1.0	2.0	5	580.0	690	none	4	2	1315	
123	3	2	1	720	4.0	420.0	1	Mac n Cheese, Lasagna, Pizza	happiness, they are some of my favorite foods	NaN	...	2.0	2.0	1	940.0	500	NaN	3	1	1315	
124	3.9	1	1	430	NaN	315.0	2	Chocolates, pizza, and Ritz.	hormones, Premenstrual syndrome.	NaN	...	1.0	2.0	2	725.0	345	NaN	4	2	575	

125 rows × 61 columns

seems like there is no duplicate in this dataset

There are some columns (19 columns) which are not related for this project. These columns are as belows,

1. breakfast
2. comfort_food
3. comfort_food_reasons
4. comfort_food_reasons_coded
5. drink
6. ethnic_food
7. father_profession
8. fries
9. greek_food
10. ideal_diet
11. indian_food
12. italian_food
13. life_rewarding
14. marital_status
15. meals_dinner_friend
16. mothers_profession
17. persian_food
18. soup
19. thai_food

After doing the overall data cleaning, from this point on, we will do the data visulization together with the related necessary data cleaning.

In [5]:
```
df = df.drop(columns = ["breakfast", "comfort_food", "comfort_food_reasons", "comfort_food_reasons_coded", "drink", "ethnic_food", "father_profession", "fries", "greek_food", "ideal_diet", "indian_food", "italian_food",
df
```

Out[5]:

	GPA	Gender	calories_chicken	calories_day	calories_scone	coffee	cook	comfort_food_reasons_coded.1	cuisine	diet_current	...	pay_meal_out	self_perception_weight	sports	tortilla_calories	turkey_calories	type_sports	veggies_day	vitamins	waffle_calories
0	2.4	2	430	NaN	315.0	1	2.0	9	NaN	eat good and exercise	...	2	3.0	1.0	1165.0	345	car racing	5	1	1315
1	3.654	1	610	3.0	420.0	2	3.0	1	1.0	I eat about three times a day with some snacks...	...	4	3.0	1.0	725.0	690	Basketball	4	2	900
2	3.3	1	720	4.0	420.0	2	1.0	1	3.0	toast and fruit for breakfast, salad for lunch...	...	3	6.0	2.0	1165.0	500	none	5	1	900
3	3.2	1	430	3.0	420.0	2	2.0	2	2.0	College diet, cheap and easy foods most nights...	...	2	5.0	2.0	725.0	690	NaN	3	1	1315
4	3.5	1	720	2.0	420.0	2	1.0	1	2.0	I try to eat healthy but often struggle becaus...	...	4	4.0	1.0	940.0	500	Softball	4	2	760
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
120	3.5	1	610	4.0	420.0	2	3.0	2	1.0	My diet consists mainly of coffee, water, frui...	...	4	4.0	1.0	940.0	500	Softball	5	1	1315
121	3	1	265	2.0	315.0	2	3.0	3	NaN	A college student with an imbalanced diet tryi...	...	4	4.0	NaN	940.0	500	basketball	5	2	1315
122	3.882	1	720	NaN	420.0	1	3.0	3	NaN	Rice, oatmeal, and tea	...	4	4.0	2.0	580.0	690	none	4	2	1315
123	3	2	720	4.0	420.0	1	3.0	7	1.0	I try to eat as healthy as possible everyday. ...	...	3	2.0	2.0	940.0	500	NaN	3	1	1315
124	3.9	1	430	NaN	315.0	2	NaN	5	3.0	high in protein and low in carbohydrates.	...	3	3.0	2.0	725.0	345	NaN	4	2	575

125 rows × 42 columns

Demographic

Let's take a look at the survey demographic in order to avoid biases towards a certain group.

In [6]:
```
selected_columns = ['Gender', 'employment', 'grade_level', 'income', 'weight']
df_selected = df[selected_columns]
df_selected.head(50)
```

Out[6]:

	Gender	employment	grade_level	income	weight
0	2	3.0	2	5.0	187
1	1	2.0	4	4.0	155
2	1	3.0	3	6.0	I'm not answering this.
3	1	3.0	4	6.0	Not sure, 240
4	1	2.0	4	6.0	190
5	1	3.0	2	1.0	190
6	2	3.0	4	4.0	180
7	1	2.0	2	5.0	137
8	1	2.0	1	5.0	180
9	1	3.0	1	4.0	125
10	1	1.0	3	3.0	116
11	1	2.0	2	5.0	110
12	1	3.0	1	5.0	264
13	2	3.0	2	5.0	123
14	2	3.0	3	5.0	185
15	2	2.0	1	4.0	180
16	2	2.0	1	1.0	145
17	2	3.0	1	6.0	170
18	1	2.0	2	5.0	135
19	1	3.0	1	5.0	165
20	2	2.0	2	6.0	175
21	2	2.0	4	6.0	195
22	2	2.0	1	6.0	185
23	2	2.0	3	4.0	185
24	1	3.0	3	4.0	105
25	1	3.0	1	6.0	125
26	2	2.0	2	5.0	160
27	2	3.0	1	6.0	175
28	2	2.0	3	5.0	180
29	2	3.0	3	5.0	167
30	1	2.0	4	5.0	115
31	2	3.0	2	4.0	205
32	2	2.0	4	5.0	NaN
33	1	3.0	1	3.0	128
34	1	3.0	1	6.0	150
35	1	3.0	3	4.0	150
36	1	2.0	2	6.0	150
37	1	2.0	4	6.0	170
38	1	3.0	4	5.0	150
39	2	3.0	4	6.0	175
40	2	3.0	1	6.0	140
41	1	2.0	2	3.0	120
42	1	2.0	2	6.0	135
43	1	NaN	3	3.0	100
44	1	2.0	4	5.0	170
45	1	2.0	2	6.0	113
46	2	2.0	1	4.0	168
47	2	NaN	1	6.0	145
48	2	3.0	2	5.0	155
49	2	NaN	3	3.0	150

After observing 50 students, **'Gender', 'grade_level' and 'income'** columns are cleaned. However, they are in numbers which is uncleared to us if we do not have access to the original CSV file.

Let's substitute these numbers with meaningful strings.

In [7]:
```
gender= {1: 'Female', 2: 'Male'}
grade_level = {1: 'Freshman', 2:'Sophomore', 3:'Junior', 4:'Senior'}
income = {1:'> $15,000', 2:'>$30,000', 3:'>$50,000', 4:'>$70,000', 5:'>$100,000', 6:'higher than $100,000'}

df['Gender'] = df['Gender'].replace(gender)
df['grade_level'] = df['grade_level'].replace(grade_level)
df['income'] = df['income'].replace(income)

selected_columns = ['Gender', 'grade_level', 'income']
df_selected = df[selected_columns]
df_selected.head(10)
```

Out[7]:

	Gender	grade_level	income
0	Male	Sophomore	>$100,000
1	Female	Senior	>$70,000
2	Female	Junior	higher than $100,000
3	Female	Senior	higher than $100,000
4	Female	Senior	higher than $100,000
5	Female	Sophomore	> $15,000
6	Male	Senior	>$70,000
7	Female	Sophomore	>$100,000
8	Female	Freshman	>$100,000
9	Female	Freshman	>$70,000

'employment' columns has some NaN values and string.

Let's replace the NaN value with 4, which is Others

Then, replacing these numbers into meaningful strings

In [8]:
```
df['employment'] = df['employment'].fillna(4).astype(int)

selected_columns = ['employment']
df_selected = df[selected_columns]
df_selected.head(50)
```

Out[8]:

	employment
0	3
1	2
2	3
3	3
4	2
5	3
6	3
7	2
8	2
9	3
10	1
11	2
12	3
13	2
14	3
15	3
16	2
17	3
18	2
19	3
20	2
21	2
22	2
23	2
24	3
25	3
26	2
27	3
28	2
29	3
30	2
31	3
32	2
33	3
34	3
35	3
36	2
37	2
38	3
39	3
40	3
41	2
42	2
43	4
44	2
45	2
46	2
47	4
48	3
49	4

In [9]:
```
df['employment'] = df['employment'].map({1: 'Full Time', 2: 'Part Time', 3: 'Not Working', 4: 'Others'})

selected_columns = ['employment']
df_selected = df[selected_columns]
df_selected.head(10)
```

Out[9]:

	employment
0	Not Working
1	Part Time
2	Not Working
3	Not Working
4	Part Time
5	Not Working
6	Not Working
7	Part Time
8	Part Time
9	Not Working

Now moved on with **weight** column

In [10]:
```
selected_columns = ['weight']
df_selected = df[selected_columns]
df_selected.head(50)
```

Out[10]:

	weight
0	187
1	155
2	I'm not answering this.
3	Not sure, 240
4	190
5	190
6	180
7	137
8	180
9	125
10	116
11	110
12	264
13	123
14	185
15	180
16	145
17	170
18	135
19	165
20	175
21	195
22	185
23	185
24	105
25	125
26	160
27	175
28	180
29	167
30	115
31	205
32	NaN
33	128
34	150
35	150
36	150
37	170
38	150
39	175
40	140
41	120
42	135
43	100
44	170
45	113
46	168
47	145
48	155
49	150

In this 'weight' column, there are some strings and NaN value.

First, replace these strings into NaN value and then replaced these NaN value with the average weight.

In [11]:
```
df['weight'] = pd.to_numeric(df['weight'], errors='coerce')

selected_columns = ['weight']
df_selected = df[selected_columns]
df_selected.head(100)
```

Out[11]:

	weight
0	187.0
1	155.0
2	NaN
3	NaN
4	190.0
...	...
95	184.0
96	210.0
97	155.0
98	165.0
99	165.0

100 rows × 1 columns

In [12]:
```
mean_weight = df['weight'].mean()
print(f"Average weight is: {mean_weight}")
```

```
Average weight is: 158.5
```

In [13]:
```
df['weight'] = pd.to_numeric(df['weight'], errors='coerce').fillna(158.5)

selected_columns = ['weight']
df_selected = df[selected_columns]
df_selected.head(100)
```

Out[13]:

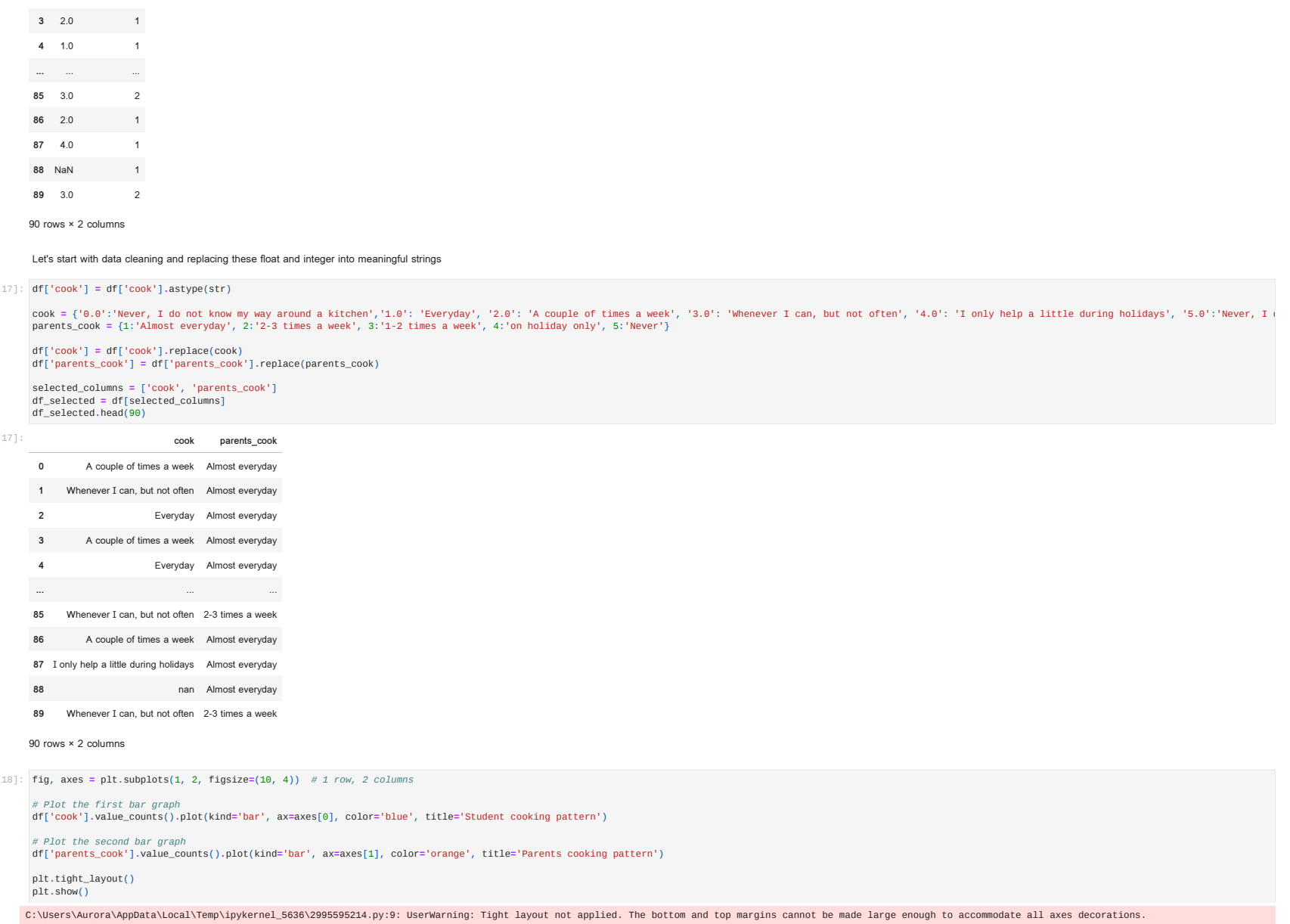
	weight
0	187.0
1	155.0
2	158.5
3	158.5
4	190.0
...	...
95	184.0
96	210.0
97	155.0
98	165.0
99	165.0

100 rows × 1 columns

After all the data cleaning, let's view the Demographic in terms of pie chart

In [14]:
```
import matplotlib.pyplot as plt
```

In [15]:
```
def create_pie_chart(column_name, subplot_index):
    plt.subplot(2, 3, subplot_index)
    column_counts = df[column_name].value_counts()
    plt.pie(column_counts, labels=column_counts.index, autopct='%1.1f%%', startangle=90)
    plt.title(f'{column_name}')

# Set up a figure with a 2x3 grid for subplots
plt.figure(figsize=(16, 12))

columns_to_plot = ['Gender', 'employment', 'grade_level', 'income', 'weight']
for index, column in enumerate(columns_to_plot, start=1):
    create_pie_chart(column, index)

plt.tight_layout()
plt.show()
```



From the above 5 pie charts, we can see this survey is done with a good mixed of demographic.

Cooking Pattern

Now we want to check whether these student cook or not based on their parents. Does their parents' habit affect on their child?

In [16]:
```
selected_columns = ['cook', 'parents_cook']
df_selected = df[selected_columns]
df_selected.head(90)
```

Out[16]:

	cook	parents_cook
0	2.0	1
1	3.0	1
2	1.0	1
3	2.0	1
4	1.0	1
...	...	...
85	3.0	2
86	2.0	1
87	4.0	1
88	NaN	1
89	3.0	2

90 rows × 2 columns

Let's start with data cleaning and replacing these float and integer into meaningful strings

In [17]:
```
df['cook'] = df['cook'].astype(str)

cook = {'6.0':'Never, I do not know my way around a kitchen','1.0': 'Everyday', '2.0': 'A couple of times a week', '3.0': 'Whenever I can, but not often', '4.0': 'I only help a little during holidays', '5.0':'Never, I d
parents_cook = {1:'Almost everyday', 2:'2-3 times a week', 3:'1-2 times a week', 4:'on holiday only', 5:'Never'}

df['cook'] = df['cook'].replace(cook)
df['parents_cook'] = df['parents_cook'].replace(parents_cook)

selected_columns = ['cook', 'parents_cook']
df_selected = df[selected_columns]
df_selected.head(90)
```

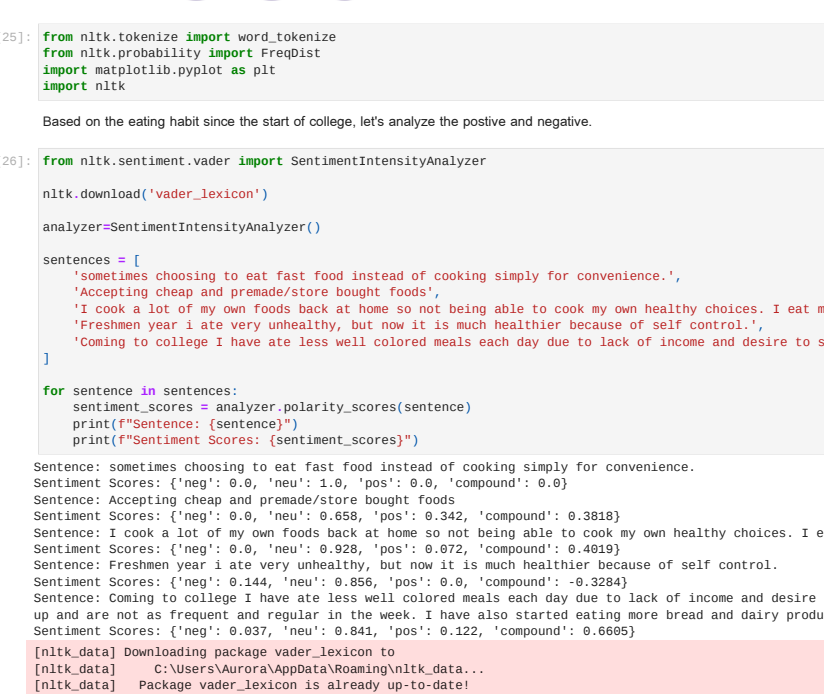
Out[17]:

	cook	parents_cook
0	A couple of times a week	Almost everyday
1	Whenever I can, but not often	Almost everyday
2	Everyday	Almost everyday
3	A couple of times a week	Almost everyday
4	Everyday	Almost everyday
...	...	...
85	Whenever I can, but not often	2-3 times a week
86	A couple of times a week	Almost everyday
87	I only help a little during holidays	Almost everyday
88	nan	Almost everyday
89	Whenever I can, but not often	2-3 times a week

90 rows × 2 columns

In [18]:
```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))  # 1 row, 2 columns

# Plot the first bar graph
df['cook'].value_counts().plot(kind='bar', ax=axes[0], color='blue', title='Student cooking pattern')

# Plot the second bar graph
df['parents_cook'].value_counts().plot(kind='bar', ax=axes[1], color='orange', title='Parents cooking pattern')

plt.tight_layout()
plt.show()
```

```
C:\Users\Aurora\AppData\Local\Temp\ipykernel_36300\2999999214.py:8: UserWarning: Tight layout not applied. The bottom and top margins cannot be made large enough to accommodate all axes decorations.
  plt.tight_layout()
```



From the above bar graph, it is very visible that parents' habit affects their children cooking pattern. Since we all know, homecook meals are way more healthier and clean than takeaways, parents who put an effort in cooking habit and make used to it can leads to their children.

However, we cannot assume this entirely. There are less than 5 parents who never cook which is relatively low in number (total is 126, so less than 5%). But more than 10 students do not cook which is double in number.

But from those who cook (both students and parents), the rest of the bar graphs show the similar number.

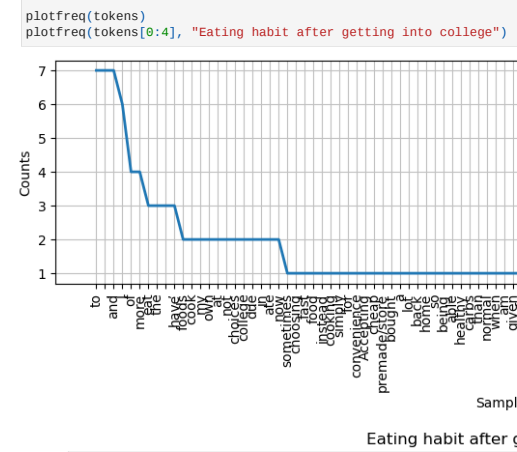
Eating Habit After Getting into College

From one of the open ended question about how is the eating habit going after getting into college, we want to analyze and visualize them in terms of words.

In [19]:
```
from wordcloud import WordCloud
```

After randomly gather 5 answers from an open ended question to display a visual narrative

In [20]:
```
sentences = [
    'sometimes choosing to eat fast food instead of cooking simply for convenience.',
    'Accepting cheap and premade/store bought foods',
    'I cook a lot of my own foods back at home so not being able to cook my own healthy choices. I eat more carbs than normal when I am at college due to the choices given in the cafe.',
    'Freshman year I ate very unhealthy, but now it is much healthier because of self control.',
    'Coming to college I have ate less well colored meals each day due to lack of income and desire to save money. I used to eat more fruits and vegetables, more chicken and salmon, now these habits have lightened up a
]

# Create and generate a word cloud image:
wordcloud = WordCloud(width=800, height=400, background_color='white').generate(full_text)

# Display the generated image:
plt.imshow(wordcloud, interpolation='bilinear')

# Bilinear interpolation is a way to spread the words out without them bumping into one another
plt.axis("off")
plt.show()
```



In [21]:
```
from nltk.tokenize import word_tokenize
from nltk.probability import FreqDist
import matplotlib.pyplot as plt
import nltk
```

Based on the eating habit since the start of college, let's analyze the positive and negative.

In [22]:
```
from nltk.sentiment.vader import SentimentIntensityAnalyzer

nltk.download('vader_lexicon')

analyzer=SentimentIntensityAnalyzer()

sentences = [
    'sometimes choosing to eat fast food instead of cooking simply for convenience.',
    'Accepting cheap and premade/store bought foods',
    'I cook a lot of my own foods back at home so not being able to cook my own healthy choices. I eat more carbs than normal when I am at college due to the choices given in the cafe.',
    'Freshman year I ate very unhealthy, but now it is much healthier because of self control.',
    'Coming to college I have ate less well colored meals each day due to lack of income and desire to save money. I used to eat more fruits and vegetables, more chicken and salmon, now these habits have lightened up a
]

for sentence in sentences:
    sentiment_scores = analyzer.polarity_scores(sentence)
    print(f"Sentence: {sentence}")
    print(f"Sentiment Scores: {sentiment_scores}")
```

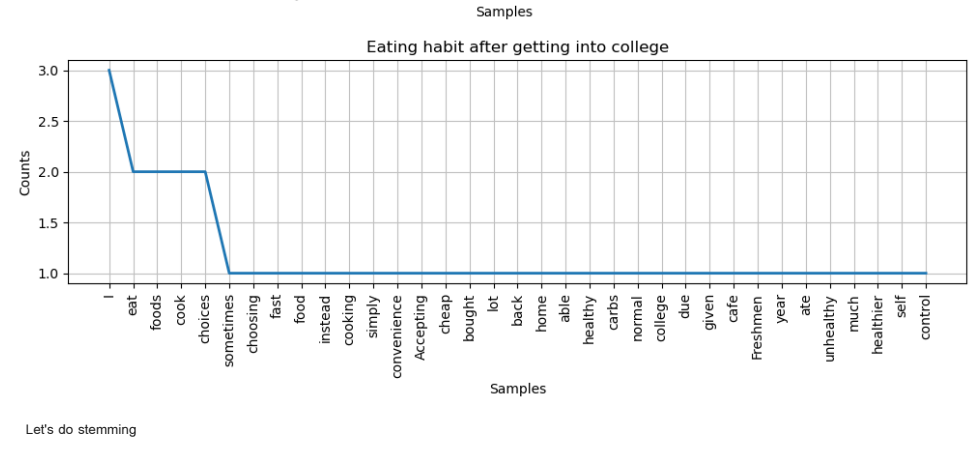
```
Sentence: sometimes choosing to eat fast food instead of cooking simply for convenience.
Sentiment Scores: {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}
Sentence: Accepting cheap and premade/store bought foods
Sentiment Scores: {'neg': 0.0, 'neu': 0.658, 'pos': 0.342, 'compound': 0.3818}
Sentence: I cook a lot of my own foods back at home so not being able to cook my own healthy choices. I eat more carbs than normal when I am at college due to the choices given in the cafe.
Sentiment Scores: {'neg': 0.0, 'neu': 0.928, 'pos': 0.072, 'compound': 0.4019}
Sentence: Freshman year I ate very unhealthy, but now it is much healthier because of self control.
Sentiment Scores: {'neg': 0.144, 'neu': 0.856, 'pos': 0.0, 'compound': -0.2284}
Sentence: Coming to college I have ate less well colored meals each day due to lack of income and desire to save money. I used to eat more fruits and vegetables, more chicken and salmon, now these habits have lightened up and are not as frequent and regular in the week. I have also started eating more bread and dairy products.
Sentiment Scores: {'neg': 0.037, 'neu': 0.841, 'pos': 0.122, 'compound': 0.4404}
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     C:\Users\Aurora\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
```

In [23]:
```
open_ended_q=[""] * 5

open_ended_q[0] = "sometimes choosing to eat fast food instead of cooking simply for convenience."
open_ended_q[1] = "Accepting cheap and premade/store bought foods"
open_ended_q[2] = "I cook a lot of my own foods back at home so not being able to cook my own healthy choices. I eat more carbs than normal when I am at college due to the choices given in the cafe."
open_ended_q[3] = "Freshman year I ate very unhealthy, but now it is much healthier because of self control."
open_ended_q[4] = "Coming to college I have ate less well colored meals each day due to lack of income and desire to save money. I used to eat more fruits and vegetables, more chicken and salmon, now these habits have

def plotFreq(x, title=""):
    flat = []
    for i in x:
        flat += i

    f = nltk.FreqDist(flat)

    # Plot freq dist
    plt.figure(figsize=(12,5))

    plt.title(title)
    f.plot()
```

In [24]:
```
tokens = []
for open_ended in open_ended_q:
    words = word_tokenize(open_ended)
    tokens.append(words)

plotFreq(tokens)
plotFreq(tokens[0:4], "Eating habit after getting into college")
```

Let's remove punctuation

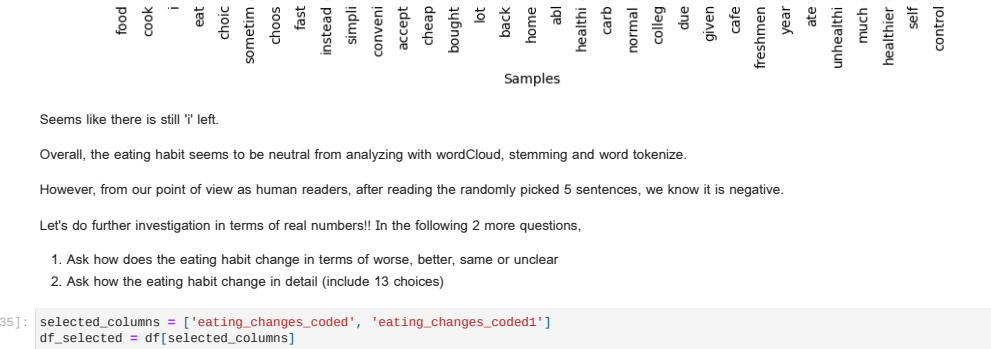
In [25]:
```
cleaned_tokens = []
for t in tokens:
    cleaned = [word for word in t if word.isalpha()]
    cleaned_tokens.append(cleaned)

plotFreq(cleaned_tokens)
plotFreq(cleaned_tokens[0:4], "Eating habit after getting into college") # dancers
```



Let's remove stopwords

In [26]:
```
from nltk.corpus import stopwords
stop_words = stopwords.words('english')
print(stop_words)

# remove stopwords
stopwords_removed_tokens = []
for t in cleaned_tokens:
    cleaned = [word for word in t if not word in stop_words]
    stopwords_removed_tokens.append(cleaned)

plotFreq(stopwords_removed_tokens)
plotFreq(stopwords_removed_tokens[0:4], "Eating habit after getting into college")
```

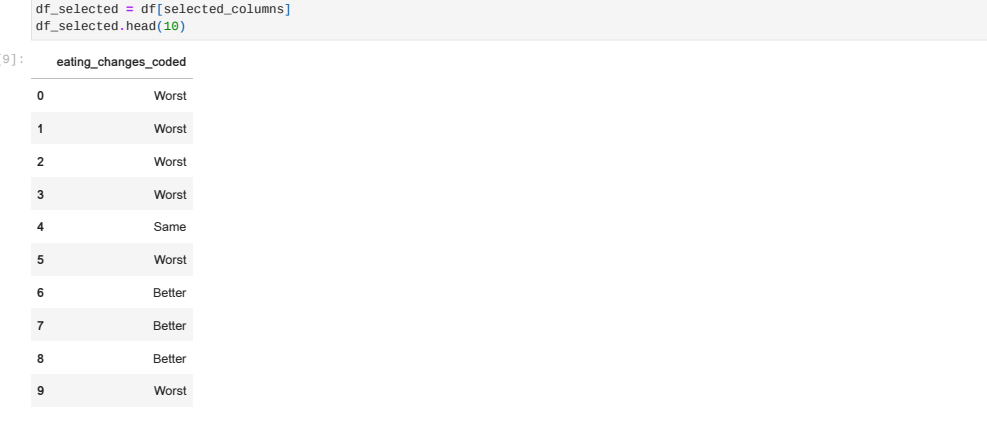
```
['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "you've", "you'll", "you'd", 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'his', 'himself', 'she', "she's", 'her', 'hers', 'herse
lf', 'it', "it's", 'its', 'itself', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'whom', 'this', 'that', "that'll", 'these', 'those', 'am', 'is', 'are', 'was', 'were', 'be', 'been', 'being',
'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing', 'a', 'an', 'the', 'and', 'but', 'if', 'or', 'because', 'as', 'until', 'while', 'of', 'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'throu
gh', 'during', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'on', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'when', 'where', 'why', 'how', 'all', 'an
y', 'both', 'each', 'few', 'more', 'most', 'other', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'too', 'very', 's', 't', 'can', 'will', 'just', 'don', "don't", 'should', "should've", 'now',
'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', "aren't", 'couldn', "couldn't", 'didn', "didn't", 'doesn', "doesn't", 'hadn', "hadn't", 'hasn', "hasn't", 'haven', "haven't", 'isn', "isn't", 'ma', 'mightn', "mightn't", 'mustn', "mustn't", 'needn', "needn't", 'shan', "shan't", 'shouldn', "shouldn't", 'wasn', "wasn't", 'weren', "weren't", 'won', "won't", 'wouldn', "wouldn't"]
```

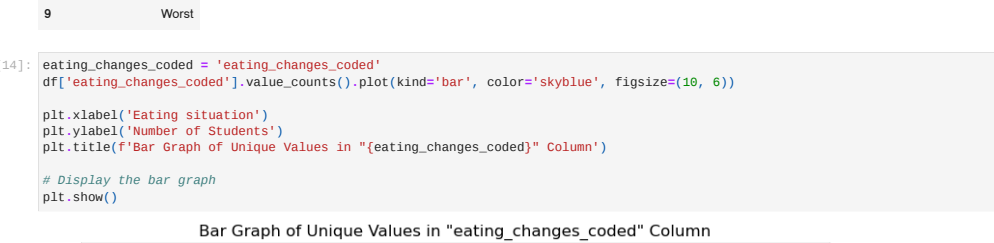


Let's do stemming

Only Porter Stemmer is used to do the mild stemming

In [32]:
```
from nltk.stem import PorterStemmer
ps = PorterStemmer()
stemmed_tokens = []
for t in stopwords_removed_tokens:
    cleaned = [ps.stem(word) for word in t]
    stemmed_tokens.append(cleaned)

plotFreq(stemmed_tokens)
plotFreq(stemmed_tokens[0:4], "Eating habit after getting into college")
```





Seems like there is still 'I' left.

Overall, the eating habit seems to be neutral from analyzing with wordCloud, stemming and word tokenize.

However, from our point of view as human readers, after reading the randomly picked 5 sentences, we know it is negative.

Let's do further investigation in terms of real numbers!! In the following 2 more questions,

1. Ask how does the eating habit change in terms of worse, better, same or unclear
2. Ask how the eating habit change in detail (include 13 choices)

In [38]:
```
selected_columns = ['eating_changes_coded', 'eating_changes_coded1']
df_selected = df[selected_columns]
df_selected.head(10)
```

Out[38]:

	eating_changes_coded	eating_changes_coded1
0	1	1
1	1	2
2	1	3
3	1	3
4	3	4
5	1	3
6	2	5
7	2	5
8	2	8
9	1	3

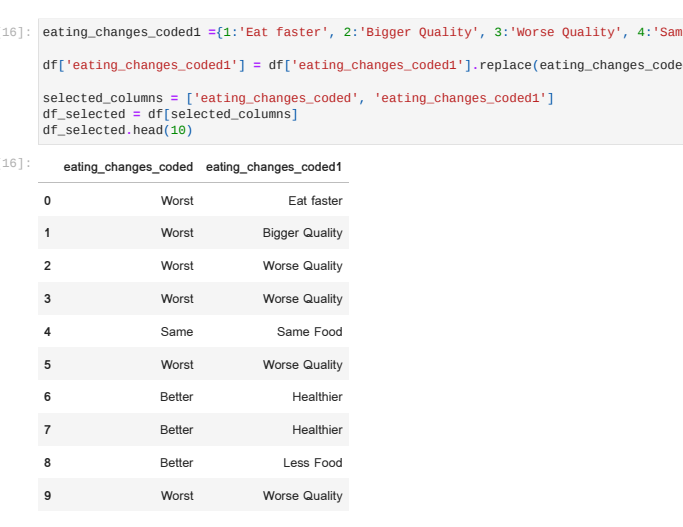
Data for these 2 columns is clean. So, let's replace these numbers with meaningful strings

In [6]:
```
eating_changes_coded ={1:'Worst', 2:'Better', 3:'Same', 4:'Unclear'}
df['eating_changes_coded'] = df['eating_changes_coded'].replace(eating_changes_coded)

selected_columns = ['eating_changes_coded']
df_selected = df[selected_columns]
df_selected.head(10)
```

Out[6]:

	eating_changes_coded
0	Worst
1	Worst
2	Worst
3	Worst
4	Same
5	Worst
6	Better
7	Better
8	Better
9	Worst

In [14]:
```
eating_changes_coded = 'eating_changes_coded'
df['eating_changes_coded'].value_counts().plot(kind='bar', color='skyblue', figsize=(10, 6))

plt.xlabel('Eating situation')
plt.ylabel('Number of Students')
plt.title(f'Bar Graph of Unique Values in "{eating_changes_coded}" Column')

# Display the bar graph
plt.show()
```

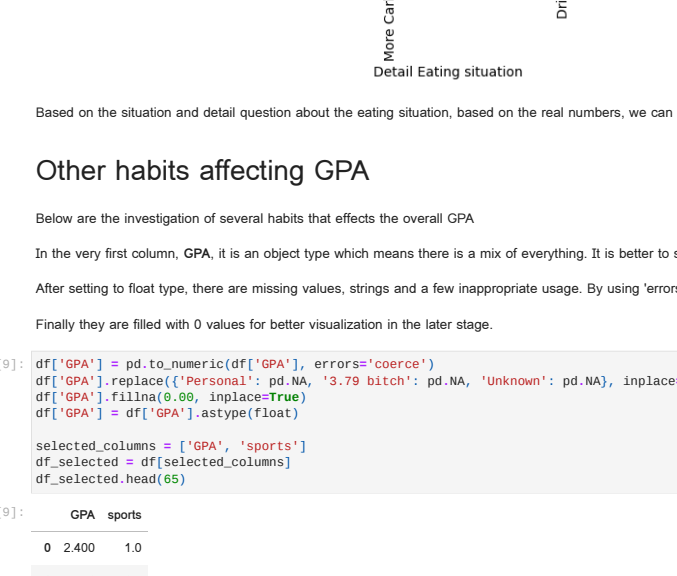


In [16]:
```
eating_changes_coded1 ={1:'Eat faster', 2:'Bigger Quality', 3:'Worse Quality', 4:'Same Food', 5:'Healthier', 6:'Unclear', 7:'Drink Coffee', 8:'Less Food', 9:'More Sweets', 10:'Timing', 11:'More Carbs or Snacking', 12:'

df['eating_changes_coded1'] = df['eating_changes_coded1'].replace(eating_changes_coded1)

selected_columns = ['eating_changes_coded', 'eating_changes_coded1']
df_selected = df[selected_columns]
df_selected.head(10)
```

Out[16]:

	eating_changes_coded	eating_changes_coded1
0	Worst	Eat faster
1	Worst	Bigger Quality
2	Worst	Worse Quality
3	Worst	Worse Quality
4	Same	Same Food
5	Worst	Worse Quality
6	Better	Healthier
7	Better	Healthier
8	Better	Less Food
9	Worst	Worse Quality

In [18]:
```
eating_changes_coded1 = 'eating_changes_coded1'
df['eating_changes_coded1'].value_counts().plot(kind='bar', color='skyblue', figsize=(10, 6))

plt.xlabel('Detail Eating situation')
plt.ylabel('Number of Students')
plt.title(f'Bar Graph of Unique Values in "{eating_changes_coded1}" Column')

# Display the bar graph
plt.show()
```



Based on the situation and detail question about the eating situation, based on the real numbers, we can clearly see that it is getting worst. Meanwhile, in the word visualization, this situation is not that visable.

Other habits affecting GPA

Below are the investigation of several habits that effects the overall GPA

In the very first column, **GPA**, it is an object type which means there is a mix of everything. It is better to set the column into float type.

After setting to float type, there are missing values, strings and a few inappropriate usage. By using 'errors="coerce"', these strings are replaced by NaN.

Finally they are filled with 0 values for better visualization in the later stage.

In [9]:
```
df['GPA'] = pd.to_numeric(df['GPA'], errors='coerce')
df['GPA'].replace({'Personal': pd.NA, '3.79 bitch': pd.NA, 'Unknown': pd.NA}, inplace=True)
df['GPA'].fillna(0.00, inplace=True)
df['GPA'] = df['GPA'].astype(float)

selected_columns = ['GPA', 'sports']
df_selected = df[selected_columns]
df_selected.head(65)
```

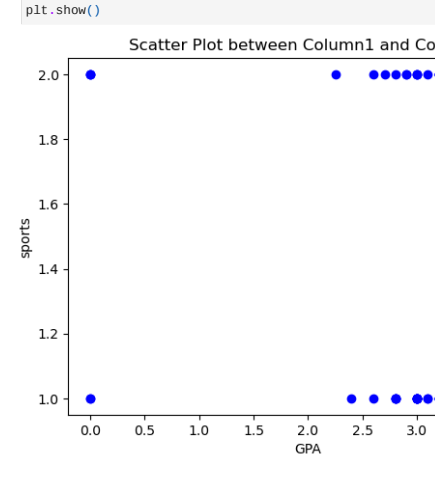
Out[9]:

	GPA	sports
0	2.400	1.0
1	3.654	1.0
2	3.300	2.0
3	3.200	2.0
4	3.500	1.0
...	...	...
60	3.600	1.0
61	0.000	2.0
62	3.900	1.0
63	2.600	1.0
64	3.500	1.0

65 rows × 2 columns

Now the data is cleaned for **GPA** column.

In [13]:
```
plt.scatter(df['GPA'], df['sports'], color='blue', marker='o')

# Adding labels and title
plt.xlabel('GPA')
plt.ylabel('sports')
plt.title('Scatter Plot between Column1 and Column2')

# Display the scatter plot
plt.show()
```



Observing from the scatter plot above, doing sports has a slight affect the overall GPA. (or) we can say those with higher GPA (above 3.0) do sport than those with lower grades (below 3.0)

1 is Doing Sports and 2 is Not doing sports

Let's check with other habits, such as **30 fruit_day**, **45 nutritional_check**, ** 57 veggies_day**

Let's change the column **GPA** datatype into integer in order to make a box plot with other habits

In [16]:
```
df['GPA'] = df['GPA'].astype(int)

selected_columns = ['GPA', 'fruit_day', 'nutritional_check', 'veggies_day']
df_selected = df[selected_columns]
df_selected.head(65)
```

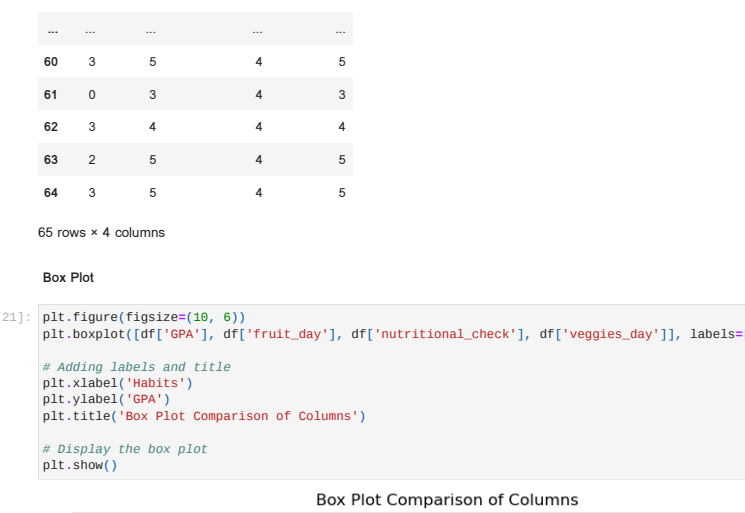
Out[16]:

	GPA	fruit_day	nutritional_check	veggies_day
0	2	5	5	5
1	3	4	4	4
2	3	5	4	5
3	3	4	2	3
4	3	4	3	4
...	...	...	...	...
60	3	5	4	5
61	0	3	4	3
62	3	4	4	4
63	2	5	4	5
64	3	5	4	5

65 rows × 4 columns

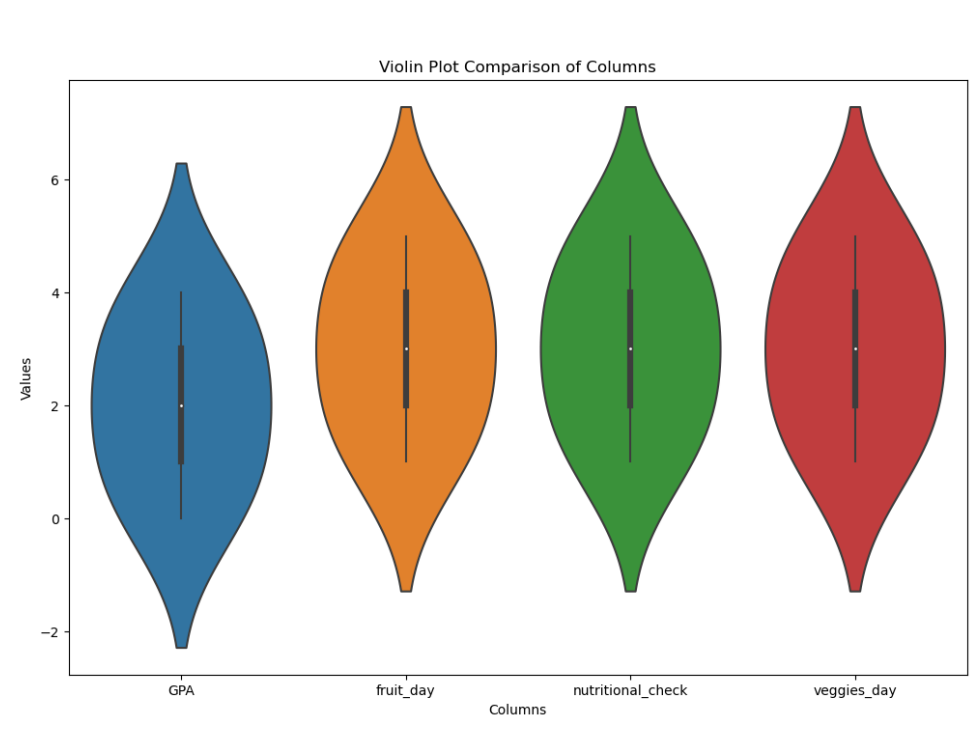
Box Plot

In [21]:
```
plt.figure(figsize=(10, 6))
plt.boxplot([df['GPA'], df['fruit_day'], df['nutritional_check'], df['veggies_day']], labels=['GPA', 'fruit_day', 'nutritional_check', 'veggies_day'])

# Adding labels and title
plt.xlabel('Habits')
plt.ylabel('GPA')
plt.title('Box Plot Comparison of Columns')

# Display the box plot
plt.show()
```



Violin Plot

In [23]:
```
import seaborn as sns

data = {'GPA': [0, 1, 2, 3, 4],
        'fruit_day': [1, 2, 3, 4, 5],
        'nutritional_check': [1, 2, 3, 4, 5],
        'veggies_day': [1, 2, 3, 4, 5]}

df = pd.DataFrame(data)

# Melt the DataFrame for better visualization
df_melted = pd.melt(df, var_name='Columns', value_name='Values')

# Violin plot
plt.figure(figsize=(12, 8))
sns.violinplot(x='Columns', y='Values', data=df_melted)

# Adding labels and title
plt.xlabel('Columns')
plt.ylabel('Values')
plt.title('Violin Plot Comparison of Columns')

# Display the violin plot
plt.show()
```



From the box plot and violin plot above, we can witness that most students scrored 3.0 GPA, regularly eating fruits and vegetables and moderately doing the nutritional check.